



**UNIVERSITY OF
MPUMALANGA**

DICT431: ICT RESEARCH METHODS

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Formalities, Psychology and Ethics

Topic Two



Lecture outline

At the end of the lecture, the student should understand:

- Formalities in the research process
- Psychology of research decisions
- The importance of research ethics and integrate research ethics into the research process.



Case study example....

You are helping your close collaborator's study in a lab. The study involves collecting blood over 6 hours in human subjects. You have made sure that you put correct labels on the blood collection tubes. When the study is over, your collaborator takes the tubes to store in a refrigerator. Two days later, you find that the labels on the tubes have been changed to make it appear that they were collected over a period of two weeks.



Case study example....

You are studying an association between sugar intake in children and obesity. Today you are recruiting a child at a local school. When interviewing the child, you discover the child speaks limited English as they grew up in the village, but seems to be interested in the study. Unfortunately, there are no translators available today and the subject does not seem to fully understand the study. Since your study does not have an invasive procedure and collects only waist circumference measurements, you wonder if you can go ahead and enrol the child.



Case study example....

While you are uploading a paper to a journal's website for submission, you get a phone call from one of your co-authors requesting to add two more names to the paper. The paper has been approved by all of your co-authors and these two people helped only enrol three more subjects in the study with 200 subjects. The requester is well-known in your field and you need his help with your post-graduate job search.



Case study example....

You have decided to submit your first abstract ever for a national conference. Preliminary data are very promising and likely to draw a lot of attention if the study is complete. Although the deadline of the abstract is just two weeks ahead, you still need 1 more healthy volunteer to complete the study. Hearing about your problem, your colleague suggests, “Why don’t you enrol yourself?”



Definition of Scientific Misconduct

Scientific misconduct is fabrication, falsification, or plagiarism in proposing, performing, or reviewing research, or in reporting research results.



Seven Areas of Scientific Dishonesty

1. Plagiarism
2. Fabrication and falsification
3. Nonpublication of data
4. Faulty data-gathering procedures
5. Poor data storage and retention
6. Misleading authorship
7. Sneaky publication practices



Plagiarism

- Plagiarism—using the ideas, writings, and drawings of others as your own



Fabrication and Falsification

- Fabrication and falsification—making up or altering data



Nonpublication of Data

- Sometimes called “cooking data”
- Data not included in results because they don’t support the desired outcome
- Some data are “bad” data
- Bad data should be recognized while it is being collected or analyzed
- Outlier – unrepresentative score; a score that lies outside of the normal scores
- How should outliers be handled?



Faulty Data Gathering

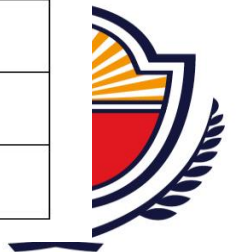
- Collecting data from participants who are not complying with requirements of the study
- Using faulty equipment
- Treating participants inappropriately
- Recording data incorrectly

The NatureMapping Program
Data Collection Form

Observer's Name/ID: _____

City/County/State: _____

Date	Species Name	Species Code	Unsure (1 or 2)	How Observed?	TRS or Lat/Long	How many?	Estimate	Habitat Code



Poor Data Storage and Retention

- Data should be stored in its original collected form for at least 3 years after publication
- Data should be available for examination
- Confidentiality of participants should be maintained



Misleading Authorship

Misleading authorship—who should be an author?

- Technicians do not necessarily become joint authors.
- Authorship should involve only those who contribute directly.
- Discuss authorship before the project!



Sneaky Publication Practices

- **Publication of the thesis or dissertation**
 - Should be regarded as the student's work?
- **Dual publication** – a manuscript should only be published in a single journal
 - What about studies which include a huge amount of data?



Class Exercise- Group

- Find an example of an unethical research study (any field) and discuss the ethical issues it raises.
- What specific research ethics principles were violated, and how could the study have been conducted ethically?



Notes

- Examples of Ethical issues in Research:

[Ethical Considerations in Research | Types & Examples
\(scribbr.com\)](https://www.scribbr.com/ethical-considerations-in-research/types-and-examples/)



RESEARCH TOPIC



STARTING THE PROCESS

1. Broad Topic
2. Preliminary Reading
3. Narrow Topic
4. Focused Topic
5. Research Question

Broad Topic

- The broadest area of your research interest.
- Your broad topic will help to guide your literature review.
- Make note of the dominant questions within the topic you are researching.
- E.g. **Democratisation in Africa.**



Narrow Topic

- After some research, you can begin to narrow the topic.
- Consider:
 - Relevant theories/models.
 - Time period.
 - Particular events.
 - Geographical area.
 - Biographical information – gender; age; ethnicity.
 - Other aspects or fields – economic; psychological; historical.
 - Available data.



Consider...

“Democratisation in Africa.”

How could you narrow this down?



Consider...

“Democratisation in Africa.”



**“The role of women in democratisation
in Anglophone Africa.”**

The broad topic was refined by using
gender and language group.

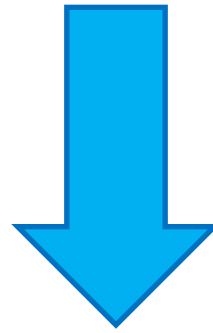


Focused Topic

- Similar process to earlier step.
- You should now be familiar with the major debates.
- You should also have a good sense of what information and resources are available.



**“The role of women in democratisation in
Anglophone Africa.”**



**“The role of urban women in democratisation in
South Africa between 1999 and 2004.”**

The Logic of Narrowing

- You should have logical reasons for narrowing the topic the way that you have.
- The reasons may vary – interest; under-researched areas; available data; word limits.
- You should make your rationale clear – this usually is covered in the “significance of study” section of your paper.



“The role of urban women in democratisation in South Africa between 1999 and 2004.”

Why 1999 and 2009?

- President Thabo Mbeki was in power for these two electoral terms.
 - This allows some control over some variables, such as leadership.
- Time since 1994, possibly allowing democratic mechanisms to be more familiar to citizens.
- Sufficient time between that period and the present for academic debates to have developed.



Research Question

- Research should allow you to realise some of the complexity of your topic.
- May have a number of sub-components to your research question.
 - These should lead to a coherent paper – do not have a series of unrelated questions for the sake of it.

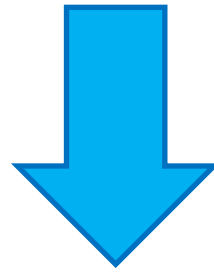


Research Question

- Think carefully before settling on a question.
- You should be able to explain the rationale behind the question set.
 - E.g. Why did you use the methodology chosen?



“The role of urban women in democratisation in South Africa between 1999 and 2004.”



“How did urban women use protest between 1999 and 2009 as a tool for promoting democracy in South Africa?”

Key Concepts

- You need to understand the key concepts in your question.
- You should define these concepts in your introduction – these often are deeply contested.



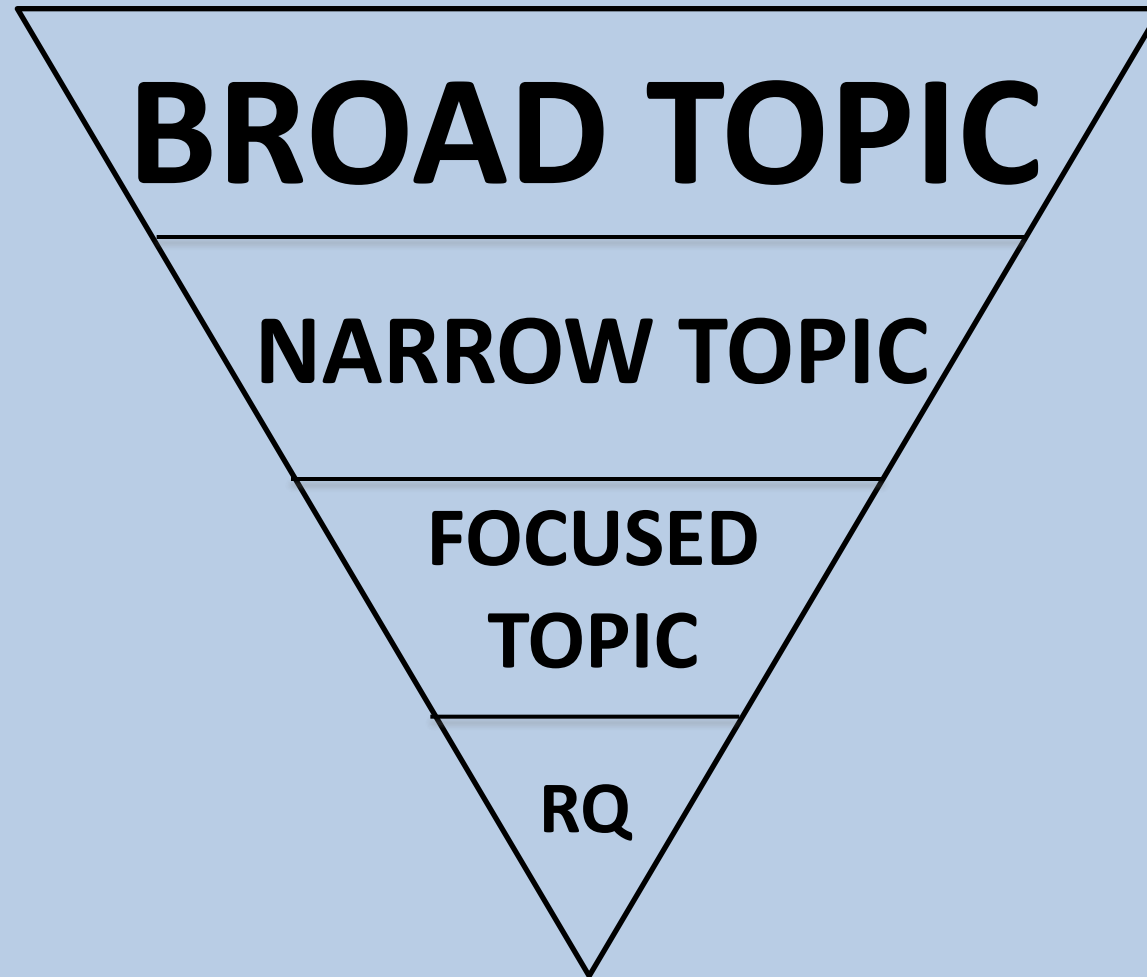
“How did urban women use protest between 1999 and 2009 as a tool for promoting democracy in South Africa?”

“Protest”; “democracy”; “promoting democracy”; “urban”; and even “tool”.

Consider “urban”

Does this include peri-urban areas? Only metropolitan areas? Only Johannesburg, Cape Town, Port Elizabeth and Durban? Why? The reason may simply be because of available and reliable data.





What to avoid

Do not...

- Underestimate time
 - Setting a question requires a lot of research, editing and reflection.
- Set a question beyond the assignment's scope
 - Usually cannot be narrow enough!
- Use unclear wording
 - Alternative phrasing?
- Feel 'stuck' with a question





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Thank You
ANY QUESTION???



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